

Scalability & Future Plan

Date	2 October 2026
Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how EduGenie can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, from a locally run learning assistant to a cloud-ready, personalized learning platform.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs locally on a single Uvicorn instance with no load balancing	Response time may slow down or requests may fail when many students use the app at the same time	High
2	No database or user accounts; no learner data is stored (local files only)	Progress tracking, study history and personalized roadmaps cannot persist across sessions	High
3	Heavy dependence on the Gemini cloud API and the local LaMini-Flan-T5-783M model, with no caching, retry or fallback	Internet or API failures break Q&A, quiz, summary and learning path; the local model is slow without a GPU	Medium
4	Limited to English text input with basic validation, and no authentication or rate limiting	Fewer students can be supported, and the API is open to misuse or overuse	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance FastAPI app run locally with Uvicorn	Containerize with Docker, run multiple Uvicorn workers behind a load balancer, and deploy on scalable cloud infrastructure with auto-scaling and health checks
2	Data Storage	No database; no user data stored, local template files only	Add a cloud database (e.g., PostgreSQL / Firestore) to store user profiles, quiz results, study plans and progress; use cloud storage for study materials
3	Performance	Direct API calls on every request; local LaMini model runs on CPU for /explain	Add response caching, async request handling, retry/fallback logic, model quantization or GPU for the local model, and latency monitoring
4	Security	API key kept in an environment variable; no user data stored	Add authentication and authorization, HTTPS, input validation, rate limiting, secret manager for keys, pinned dependencies and vulnerability scanning

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve existing features: better error handling, input validation, adjustable quiz length/level, sample questions and step-by-step explanations	1 - 2 Months	Better usability, more accurate answers and a smoother user experience
Phase 2	Add user login, database and a progress dashboard with goal tracking; diagnostic gap checks and personalized study plans	3 - 4 Months	Students can track progress, identify learning gaps and follow a personalized roadmap
Phase 3	Cloud deployment with Docker, load balancer, caching and monitoring; mobile-friendly app and multi-language support	5 - 6 Months	Supports more students reliably with faster responses and wider reach
Phase 4	Multiple subjects and grades, text/example/visual explanations, school/college platform integration and freemium / institutional plans	7 - 9 Months	Scales EduGenie into a full adaptive learning platform for institutions